

Muhammad Ahmad

AI / ML Engineer

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SUMMARY

AI/ML engineer with hands-on production experience shipping autonomous agent systems, real-time voice AI, and ML classifiers end to end. Built and deployed 7+ production systems, including a voice agent at sub-1-second latency and a text classifier reaching 100% task recall at 12.7ms inference. Strong backend foundation (FastAPI, multi-database systems) paired with a rigorous, metrics-driven ML experimentation practice.

EXPERIENCE

VaconAI

Sept 2025 – Aug 2026

AI/ML Engineer, Contract

Remote

- Built a knowledge-graph service spanning three databases (Neo4j, MongoDB, Pinecone; 6 node types, 15 relationship types) that converts legal deposition transcripts into queryable contradiction and admission graphs in under 2 minutes end to end.
- Automated PayPal dispute resolution for an e-commerce payments platform with a 10-agent LLM routing system and a 4-tier offer-escalation ladder, replacing manual case handling with a bounded, trackable negotiation flow.
- Designed and shipped a small-talk filter for voice agents, running an 11-track ML experimentation process from a baseline logistic-regression model through to a distilled cross-encoder.
- Reached 100% task recall and 92.9% chitchat rejection at 12.7ms inference latency, meeting a strict recall-floor constraint required for production voice-agent deployment.

Arbisoft

July 2024 – June 2025

Software Developer

Remote

- Built a six-phase autonomous content pipeline (keyword research, pillar/cluster mapping, live-ranking content briefs, generation, on-page optimization) orchestrated through the Claude Code CLI in headless mode, publishing 364+ verified-live articles across 13 sites.
- Replaced two separate hardcoded Python systems (research/drafting and video rendering) with a single Discord-controlled agent that reads its own operating rules from markdown files instead of code, turning most new requirements into a documentation edit instead of a deploy.
- Built a quality-scoring gate (numeric threshold plus a banned-phrase filter for AI-sounding language) that ships strong drafts automatically, revises borderline ones, and discards weak ones, behind a hard human-approval step before anything publishes.
- Shipped 364+ articles and drove 14,000+ views across five YouTube channels over five months of unattended operation, with retry logic that carried the pipeline through transient failures and resumed automatically after usage-cap resets.

Techomnis

Feb 2023 – Nov 2023

Software Developer

Remote

- Built the core of a medical transport management system handling 500+ daily trips, covering trip scheduling, status tracking, and client record management.
- Added a reporting dashboard and data-export tooling on top of the trip-management system, giving staff visibility into daily and historical operations.
- Built a CSV bulk-import pipeline with validation for onboarding client and trip data, replacing manual row-by-row entry.
- Implemented backend APIs with advanced filtering, pagination, and aggregation for client management, and delivered a private production deployment with invite-only access control.

TECHNICAL SKILLS

Languages: Python, TypeScript, SQL, C++

AI/ML & Agents: OpenAI Agents SDK, LLM Orchestration, RAG, MCP, PyTorch, Scikit-learn, LoRA Fine-Tuning, Cross-Encoder/SetFit Distillation, NLP

Backend & Infra: FastAPI, Flask, Docker, Modal, AWS, Google Cloud Platform, Async/Concurrent Processing

Data & Storage: Neo4j, MongoDB, Pinecone, PostgreSQL, SQLite FTS5, Apache Spark

Real-Time & Voice: Pipecat, LiveKit, WebRTC, Twilio, Deepgram, Whisper

Tools: Git, Linux, CI/CD, Streamlit

EDUCATION

Punjab University College of Information Technology (PUCIT)

2020 – 2024

B.S. Computer Science

Lahore, Pakistan

- Key coursework: Machine Learning, Data Structures, Database Systems, Statistics, Computer Vision.